PROJECT PRESENTATION

# Medical Visual Question Answering for Chest X-rays

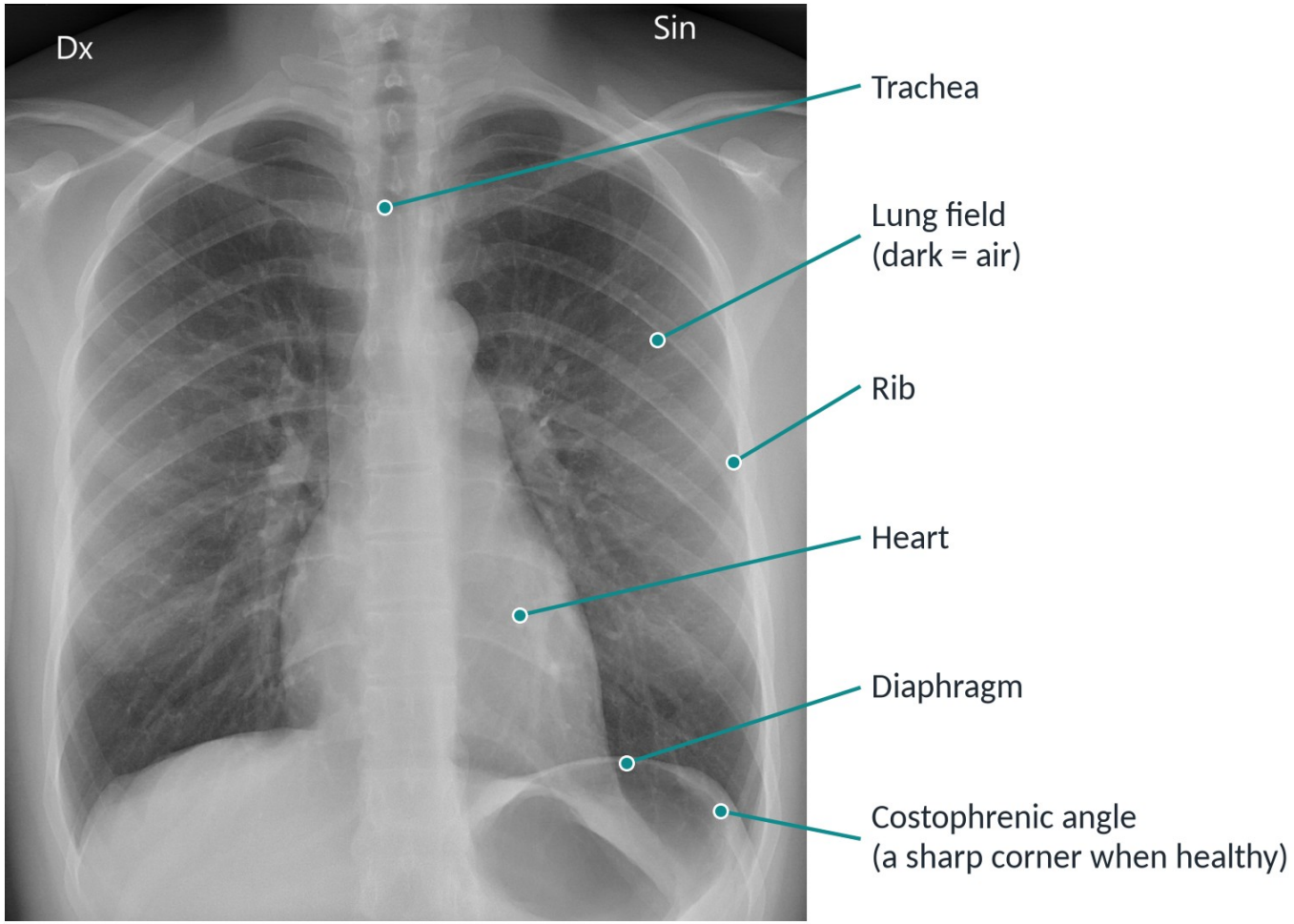
A knowledge-grounded approach with CheXagent-2-3b: literature review, baseline tests and retrieval-augmented design

**Supervisor** Dr. Snober Mushtaq

**Students** Ankrit (Enrollment 2023BCSE028) Hirdyansh (Enrollment 2023BCSE036)

Department of Computer Science and Engineering

# A chest X-ray is a shadow picture: dark means air, white means dense tissue




Four findings used in this project

## Cardiomegaly

Heart wider than half the chest

## Pleural effusion

Fluid blunts the corner at the lung base

## Pneumothorax

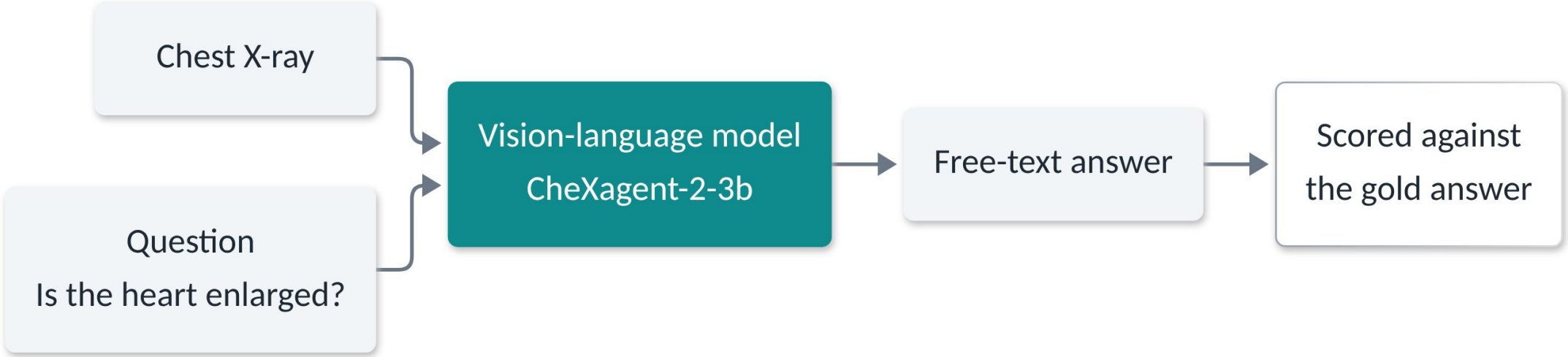
Dark area with no lung markings

## Atelectasis

Collapsed lung looks whiter

Photograph: M. Haggstrom, Wikimedia Commons, CC0 (normal PA chest X-ray). Findings: standard radiology teaching material.

# Medical VQA: show the model an X-ray and a question, then check its free-text answer




## Perception

Can the model see the finding?

Is there an effusion? Which side? Is the heart enlarged?

## Knowledge

Does the model know the disease?

What causes it? What are the symptoms? How is it treated?

Datasets: SLAKE and VQA-RAD, chest X-ray questions only. Model: StanfordAIMI/CheXagent-2-3b.

# Of six papers reviewed, three shaped the project; none combines retrieval with chest X-rays

| | Chest X-ray focus | Free-text answers | Uses retrieval | Clinician / blind rating | Textbook source | what we took from it |
|---|---|---|---|---|---|---|
| **CheXagent (2024)** | ● | ● | ○ | ● | ○ | our baseline model; weakest on left/right and severity (77-79%) |
| **RULE (2024)** | ○ | ● | ● | ○ | ○ | naive retrieval hurts: 47-59% of its errors follow the text |
| **Med-Flamingo (2023)** | ○ | ● | ○ | ● | ○ | automated scores disagree with clinicians: rate blind |
| LLaVA-Med (2023) | ○ | ● | ○ | ○ | ○ | cheap tuning; GPT-4-written training data |
| PMC-VQA (2023) | ○ | ● | ○ | ○ | ○ | caption-made questions can be answered from text alone |
| PubMedCLIP (2023) | ○ | ○ | ○ | ○ | ○ | the image encoder matters more than the text encoder |
| **This project** | ● | ● | ● | ○ | ● | **none of the six papers combines these** |

**CheXagent** is our baseline, RULE warns about retrieval, Med-Flamingo says to rate answers by hand.

Chen et al. 2024 (CheXagent); Xia et al. 2024 (RULE); Moor et al. 2023 (Med-Flamingo); Eslami et al. 2023 (PubMedCLIP); Li et al. 2023 (LLaVA-Med); Zhang et al. 2023 (PMC-VQA).

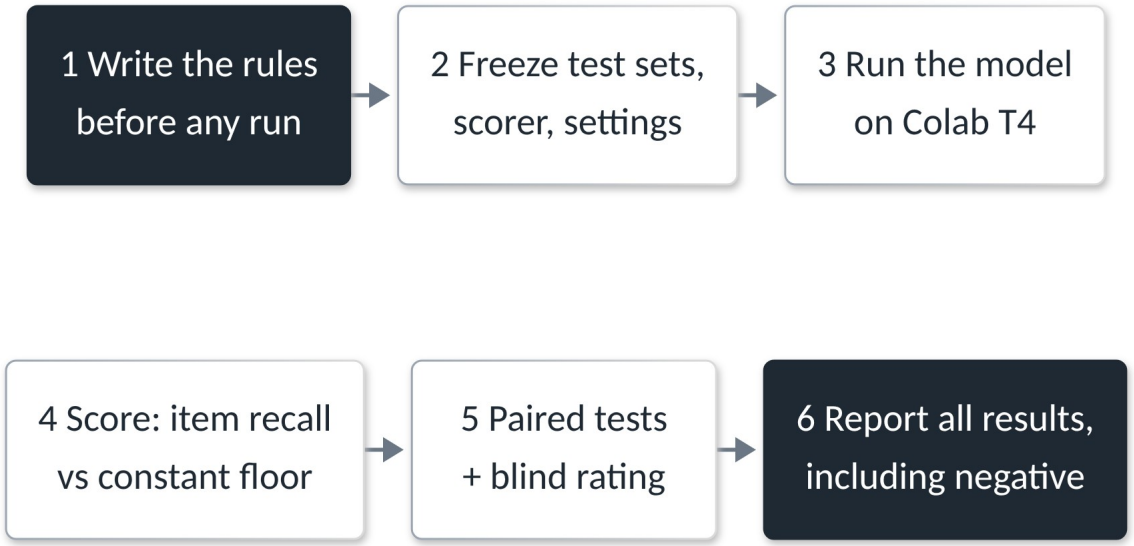
# Each finding in the literature became a concrete design decision

| What the papers show | What we decided to do |
| --- | --- |
| CheXagent is weakest on left/right and severity | Test perception alone: mirror-image and heart-size tests |
| Naive retrieval makes answers worse (RULE) | Short passages, no-harm check, disease-name-only arm |
| Automatic scores disagree with clinicians | Blind 0-1-2 rating and a constant-answer floor |
| No reviewed paper retrieves textbook knowledge | BM25 over MedlinePlus and medical textbooks |

**Research question:** can retrieved textbook knowledge make a chest X-ray model answer knowledge questions better?

CheXagent: 77-79% on fine-grained reasoning. RULE: naive retrieval over-trust. Med-Flamingo: clinician rating.

# Rules are written before any run, and negative results are reported




**219**
baseline questions

**130**
knowledge questions

**140**
left / right questions

**120**
mirror-image pairs

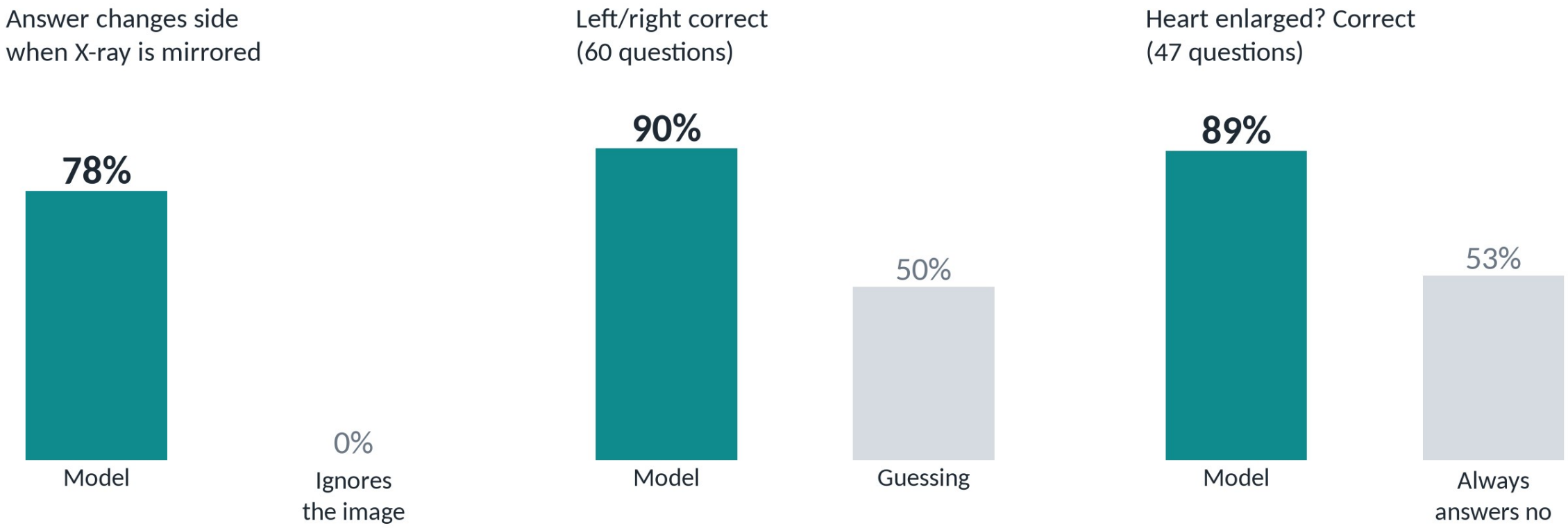
**47**
heart-size questions

**130**
images with a disease label

**Pre-registration:** test sets, scorer and success conditions were frozen first, then amended in writing when needed.

docs/evaluation_protocol.md and docs/phase2_rag_plan.md. Scorer: item recall v2; floor: most common gold answer.

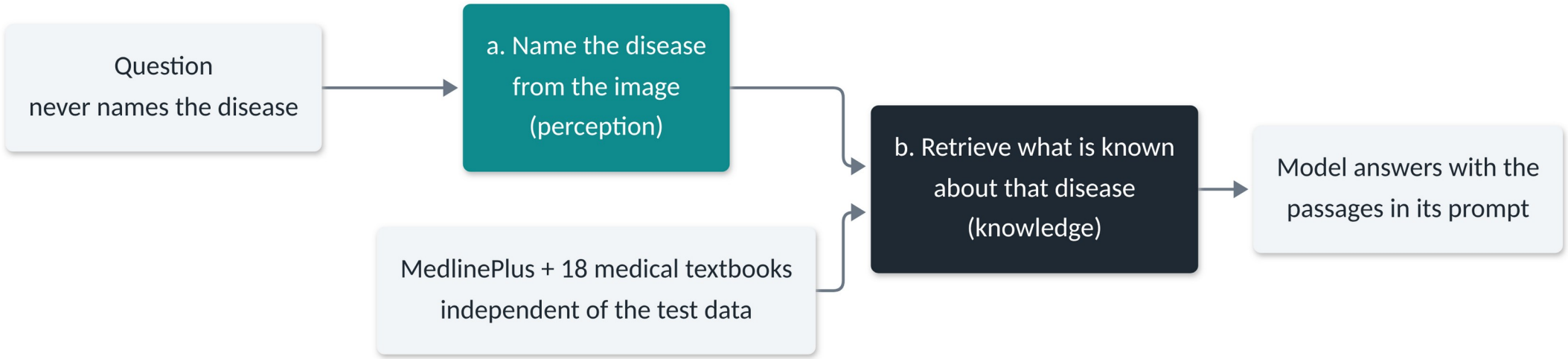
# Reading the X-ray: the model's left/right and heart-size answers follow the image




**Mirroring the X-ray flips the answer in 78% of pairs (95% CI 69-85%).** Left/right 90% correct; enlarged hearts found in 17 of 22, normal hearts called normal 25 of 25.

Colab T4, 13 Nov 2026, notebook 03, 788 generations. Data: results/run_6_with_image/. Plain wording; 120 mirror pairs, 60 side questions, 47 heart questions.

# Phase 2 design: first name the disease from the image, then retrieve what is known about it




## Why two steps: the answer depends only on the disease

| Disease on the image | Question about | Gold answer, on every such image |
|---|---|---|
| **Pneumonia** | cause | Bacterial infection |
| **Pneumonia** | treatment | Medical therapy, supportive therapy |
| **Cardiomegaly** | cause | Hypertension, dilated cardiomyopathy |

Corpora: MedlinePlus (US National Library of Medicine, public domain) and medical textbooks (MedRAG); our own BM25 search.

# Six arms separate what retrieval adds from what knowing the disease adds

| Arm | Disease named | Retrieved text | Role |
|---|---|---|---|
| **A1** | none | none | fair baseline, same answer style |
| **A2** | none | from the question alone | oracle subset: why the disease matters |
| **A3** | true disease (oracle) | none | upper bound: value of knowing the disease |
| **A4** | true disease (oracle) | yes | upper bound: value of retrieval |
| **A5** | model's own guess | none | real: the guess alone |
| **A6** | model's own guess | yes | the system |

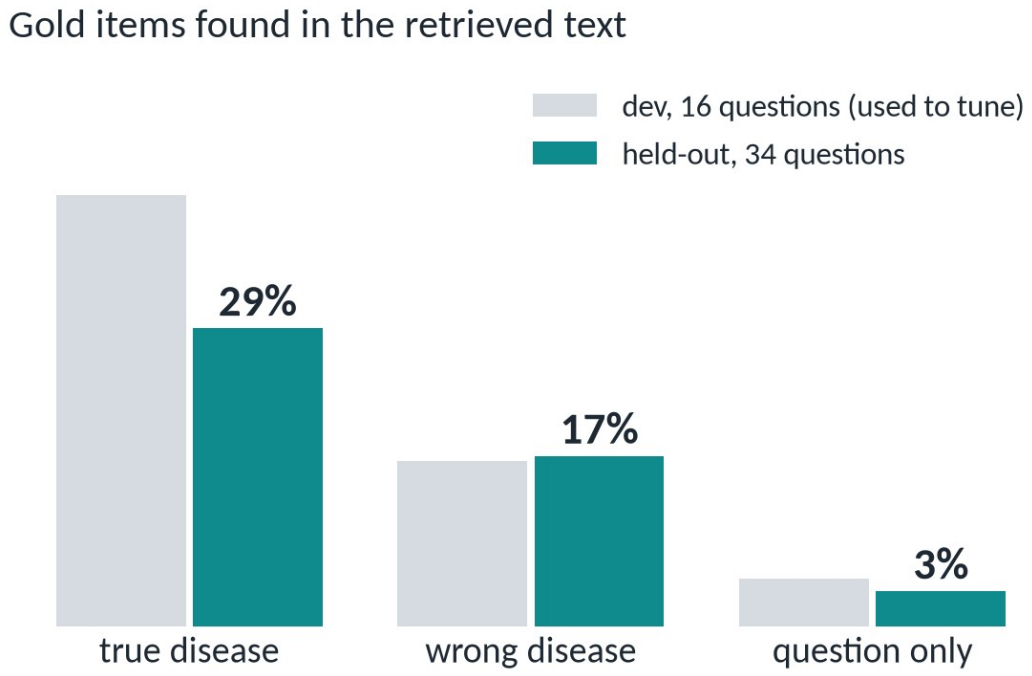
oracle: uses SLAKE's own label, an upper bound, never the result

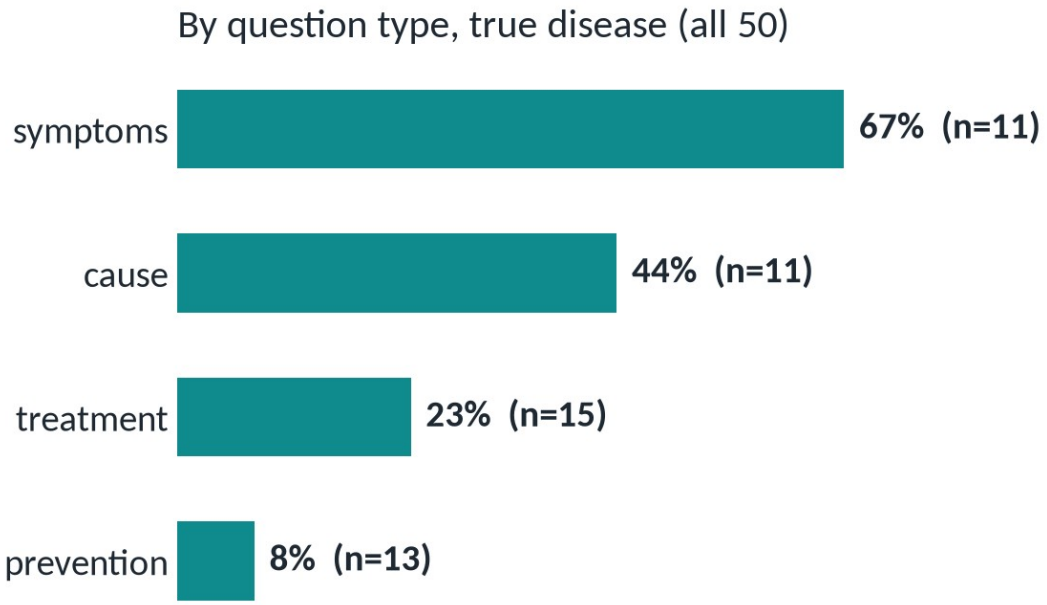
real: uses only what the model outputs

**Success needs more than a score gain:** the system (A6) must beat both the fair baseline (A1) and the guess alone (A5).

docs/phase2_rag_plan.md and Amendment 5 of the evaluation protocol.

# Retrieval check, run without any model: the corpus holds the answer mainly when the disease is right




**Held-out:** 29% of gold items found with the true disease, 17% with a wrong disease, 3% from the question alone.

Strict coverage: gold item words inside one sentence of the retrieved text. 50 oracle-evaluable questions; setting frozen on 16 dev questions.

# References


[1] Chen Z., Varma M., Delbrouck J.-B., et al. CheXagent: Towards a Foundation Model for Chest X-Ray Interpretation. arXiv:2401.12208, 2024.

[2] Xia P., Zhu K., Li H., et al. RULE: Reliable Multimodal RAG for Factuality in Medical Vision Language Models. EMNLP, 2024.

[3] Moor M., Huang Q., Wu S., et al. Med-Flamingo: a Multimodal Medical Few-shot Learner. ML4H, PMLR, 2023.

[4] Eslami S., Meinel C., de Melo G. PubMedCLIP: How Much Does CLIP Benefit VQA in the Medical Domain? Findings of EACL, 2023.

[5] Li C., Wong C., Zhang S., et al. LLaVA-Med: Training a Large Language-and-Vision Assistant for Biomedicine in One Day. NeurIPS D&B, 2023.

[6] Zhang X., Wu C., Zhao Z., et al. PMC-VQA: Visual Instruction Tuning for Medical Visual Question Answering. arXiv:2305.10415, 2023.

[7] Liu B., Zhan L.-M., Xu L., et al. SLAKE: A Semantically-Labeled Knowledge-Enhanced Dataset for Medical VQA. IEEE ISBI, 2021.

[8] Lau J. J., Gayen S., Ben Abacha A., Demner-Fushman D. A dataset of clinically generated visual questions and answers about radiology images. Scientific Data, 2018.

[9] Xiong G., Jin Q., Lu Z., Zhang A. Benchmarking Retrieval-Augmented Generation for Medicine (MedRAG). Findings of ACL, 2024.

[10] Robertson S., Zaragoza H. The Probabilistic Relevance Framework: BM25 and Beyond. Foundations and Trends in IR, 2009.

[11] Haggstrom M. Normal posteroanterior (PA) chest radiograph (X-ray). Wikimedia Commons, CC0.